LIBBY — MANUSCRIPTS INVENTORY

Manuscripts considered — pancreatic-recurrent-kras-g12r-m8f3

Master inventory: every paper reviewed in this case

Generated 2026-07-10

Manuscripts considered — pancreatic-recurrent-kras-g12r-m8f3

Master inventory: 23 clinical (19 included, 4 considered & excluded) + 22 pre-clinical (19 included, 3 considered & excluded) + 28 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Wolpin/Hong (2026) — New England Journal of Medicine

Reference:PMID 42090791 · **Type:** clinical · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236)
- **Indication / model:** Previously treated advanced RAS-mutated pancreatic ductal adenocarcinoma (PDAC) (2L+); RMC-6236-001 (NCT05379985) phase 1/2 PDAC cohort; 168 PDAC patients dosed at 300 mg or lower; second-line RAS G12 subgroup analyzed at the 300-mg dose (n=26). G12X variants enrolled include G12D, G12V, G12R, and Q61H — the patient's G12R is in scope.
- **Design:** Open-label multicenter phase 1/2 dose-escalation and dose-expansion
- **n:** 168
- **Effect size:** 35 % — ORR (RECIST 1.1), median PFS, median OS, median DoR in 2L RAS G12 PDAC at 300 mg
- **Toxicities:** any treatment-related AE (n=168, 96%); any treatment-related AE G>=3 (n=168, 30%); rash (n=127, 91%); rash G>=3 (n=127, 8%); diarrhea (n=127, 48%); diarrhea G>=3 (n=127, 2%); nausea (n=127, 43%); stomatitis (n=127, 31%); stomatitis G>=3 (n=127, 3%); vomiting (n=127, 31%); fatigue (n=127, 20%)
- **Case fit:** strong
- **Notes:** Pivotal NEJM publication of the RMC-6236-001 PDAC cohort. The ORR 35% / mPFS 8.5 / mOS 13.1 in 2L RAS G12 at 300 mg is the load-bearing efficacy anchor for this case — G12R sits inside the G12 pooled cohort. G12R-specific subgroup ORR was not separately published; the field reads G12X as a single estimand pending phase 3 readout. Allele-resolved breakdown is the open question.

Erasca / Joyo Pharmatech/— (2026)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** JYP0015 / ERAS-0015 (pan-RAS molecular glue)
- **Indication / model:** KRAS G12X-mutant advanced solid tumors — NSCLC and PDAC cohorts (2L+); KRAS G12X PDAC (2L) and NSCLC (2L+, including post-ICI/platinum); G12R covered under the G12X gate rather than named as a separate cohort
- **Design:** Phase 1 dose-escalation (AURORAS-1 US, JYP0015M101 China), open-label single-arm
- **n:** 20
- **Effect size:** 40 % — unconfirmed ORR (PDAC, RECIST 1.1)
- **Toxicities:** dose-limiting toxicity G>=3 (0%)
- **Case fit:** partial
- **Notes:** Data from Erasca 27 Apr 2026 topline release; no peer-reviewed publication or abstract yet, so per-term grade-3+ AE rates and confirmed responses are unavailable. G12R is not separately reported but falls within the G12X activity and ctDNA-clearance signal — predictive confidence for this specific allele is lower than for the pooled G12X estimate.

Kestrel Therapeutics/— (2026)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** KST-6051 / FALCON (pan-KRAS ON/OFF inhibitor)
- **Indication / model:** Advanced / metastatic KRAS-mutant solid tumors — PDAC, CRC, NSCLC named (2L+); Documented KRAS mutation without an allele-restricting list; G12R within scope. Prior KRAS/RAS inhibitor excluded
- **Design:** Phase 1 first-in-human dose-escalation (FALCON)
- **Effect size:** — — safety / RP2D (ongoing)
- **Toxicities:** No clinical safety data available; trial enrolling at dose-escalation.
- **Case fit:** strong
- **Notes:** First patient dosed 22 Apr 2026 (Kestrel/GlobeNewswire); no efficacy or safety readout exists yet, so toxicities[] is empty because no source AE data has been generated. Switch-II-pocket ON/OFF pan-KRAS mechanism; PDAC is a named enrolling tumor type.

Huff/Zaidi (2026) — Nature Communications

Reference:PMID 41667470 · **Type:** clinical · **Inclusion:** included

- **Intervention:** mutant-KRAS peptide vaccine + dual checkpoint (evidence anchor for NCT06411691)
- **Indication / model:** Resected PDAC (adjuvant); parent platform for the metastatic MMR-p PDAC/CRC trial NCT06411691 (adj); Resected PDAC; six-mutation mKRAS panel including G12R. n=12. The metastatic NCT06411691 (mKRASvax + balstilimab/botensilimab) is trial-in-progress with no efficacy readout
- **Design:** Phase 1 open-label single-arm (NCT04117087)
- **n:** 12
- **Effect size:** 11/12 with increased mKRAS T-cell response — T-cell immunogenicity
- **Toxicities:** immune-related AE G3 (2/12, 16.7%); injection-site pain (10/12, 83.3%); fatigue (4/12, 33.3%); fever (4/12, 33.3%); erythema (4/12, 33.3%)
- **Case fit:** partial
- **Notes:** Enrollable trial NCT06411691 is a metastatic MMR-p PDAC/CRC TPS (ASCO 2025 TPS2698) with no efficacy readout; this resected-PDAC phase I from the same Hopkins vaccine platform is the closest published evidence and its 6-mutation panel includes G12R. Different line (adjuvant vs the patient's recurrent disease) and dual checkpoint uses ipi/nivo rather than the trial's balstilimab/botensilimab.

BMS/BMS (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** BMS-986504 (navlimetostat / PRMT5 inhibitor) ± daraxonrasib, immunotherapy, or chemotherapy
- **Indication / model:** Solid tumors (GBM, melanoma, NSCLC, PDAC) with homozygous MTAP deletion (2L+); Homozygous MTAP deletion
- **Design:** Phase 2 platform combinations (MountainTAP-5)
- **Effect size:** — — ORR, PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Kestrel Therapeutics/Kestrel Therapeutics (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** KST-6051 oral pan-KRAS (ON/OFF) inhibitor
- **Indication / model:** Advanced solid tumors with KRAS mutation — PDAC, NSCLC, CRC named (2L+); KRAS mutation (pan-KRAS ON/OFF; G12R within scope)
- **Design:** Phase 1 dose-escalation (FALCON)
- **Effect size:** — — Safety, RP2D, ORR
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Wainberg/O'Reilly (2025) — Nature Medicine

Reference:PMID 40790272 · **Type:** clinical · **Inclusion:** included

- **Intervention:** ELI-002 2P (amphiphile mKRAS peptide vaccine) — final AMPLIFY-201 report
- **Indication / model:** Resected PDAC and CRC with MRD positivity post locoregional therapy (AMPLIFY-201 final follow-up) (adj); Final report at 19.7 months median follow-up; n=25 (20 PDAC, 5 CRC); KRAS G12D / G12R targeted.
- **Design:** Phase 1 open-label single-arm
- **n:** 25
- **Effect size:** — — Final RFS, OS, sustained T-cell responses
- **Toxicities:** Maintained tolerability at extended follow-up; 71% of evaluable patients induced both CD4 and CD8 KRAS-specific T-cell responses.
- **Case fit:** partial
- **Notes:** Final 2025 readout. Strengthens the vaccine class signal in MRD-positive resected disease and supports the ongoing 7-peptide successor (ELI-002 7P, NCT05726864). Overt recurrence sits outside the trial's enrollment window for this case. — (primary endpoint not extracted)

Banerjee/Banerjee (2025) — Journal of Clinical Oncology

Reference:PMID 40644648 · **Type:** clinical · **Inclusion:** included

- **Intervention:** avutometinib (RAF/MEK clamp) + defactinib (FAK inhibitor)
- **Indication / model:** Recurrent low-grade serous ovarian cancer (LGSOC), KRAS-mutant and wild-type (RAMP 201) (2L+); Recurrent LGSOC after ≥1 prior systemic therapy; KRAS-mutant ORR pulled separately. Cross-tumor: not PDAC but the load-bearing efficacy precedent for the RAF/MEK + FAK clamp combo in a KRAS-mutant disease.
- **Design:** Phase 2 multicenter open-label
- **Effect size:** 44 % ORR in KRAS-mutant LGSOC — Confirmed ORR by BICR; mPFS; mDoR; subset analyses by KRAS status
- **Toxicities:** Manageable AE profile; most patients continued therapy. Class effects include rash, edema, CK elevation, and reversible ocular events.
- **Case fit:** cross_tumor_only
- **Notes:** FDA accelerated approval for KRAS-mutant recurrent LGSOC based on this readout. Cross-tumor precedent for the avutometinib+defactinib backbone that RAMP-205 (NCT05669482) is testing in 1L PDAC.

O'Hara/Flaherty (2025) — Clinical Cancer Research

Reference:PMID 39437014 · **Type:** clinical · **Inclusion:** included

- **Intervention:** palbociclib (PD-0332991)
- **Indication / model:** Histology-agnostic advanced solid tumors with CDK4 or CDK6 amplification (NCI-MATCH subprotocol Z1C) (2L+); 38 eligible/treated patients with confirmed CDK4 or CDK6 amplification (≥7 copies). 25 patients in the primary analysis cohort. Pancreatic cancer was not the dominant histology; the read-out is for the CDK4/6 amplification call, not CDKN2A loss.
- **Design:** Phase 2 basket trial arm (single-arm)
- **n:** 38
- **Effect size:** 4 % ORR — ORR (primary), mPFS, mOS
- **Toxicities:** neutropenia G>=3 (n=38)
- **Case fit:** weak
- **Notes:** Negative basket readout for palbociclib in CDK4/6-amplified tumors (ORR 4%). This is the closest published precedent for off-label palbociclib in a CDKN2A-loss / CCND3 case — the message it sends is that monotherapy CDK4/6 inhibition does not deliver in unselected biomarker-positive solid tumors. CDKN2A-loss enrichment (the patient's actual feature) was not the gating biomarker here; CAPTUR Group 8 and NCI-MATCH Z1F are the more relevant arms but those readouts are not yet published.

Jacobio Pharmaceuticals/— (2025)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** JAB-23E73 (pan-KRAS ON/OFF inhibitor)
- **Indication / model:** Advanced solid tumors harboring KRAS gene alteration (2L+); KRAS-altered solid tumors including G12X (G12D/V/R/S/A), G13D, Q61H; spares HRAS/NRAS. RAS-inhibitor-naive
- **Design:** Phase 1/2a open-label dose-escalation (first-in-human, US and China)
- **Effect size:** — — safety / RP2D (ongoing)
- **Toxicities:** No clinical safety data released. Preclinical models showed tumor regression without significant body-weight loss.
- **Case fit:** partial
- **Notes:** AACR-NCI-EORTC 2025 disclosure is preclinical; the phase 1 is first-in-human with only a qualitative 'early antitumor activity' mention (Citeline/Scrip), no numeric efficacy or per-term AE data. toxicities[] left empty because no clinical AE table exists — abstract/registry carry no safety tab yet. G12R named among covered G12X alleles.

Singhi/Bahary (2025) — Cancer Research

Reference:doi:10.1158/0008-5472.CAN-25-0428 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** KRAS G12R-mutant PDAC distinctive features (ERK / MAPK and tumor microenvironment)
- **Indication / model:** Retrospective molecular and transcriptomic analysis of 2,447 PDAC tumors with paired organoid and mouse-model validation (KrasG12R vs KrasG12D ductal organoids; orthotopic implants)
- **Design:** G12R produces a weaker ERK / MAPK transcriptional output than other G12 alleles, with reduced downstream KRAS-signature activation. The functional consequence is increased relative dependence on residual MAPK signaling and a distinctive immune-stromal microenvironment that may behave differently under direct RAS-GTP blockade.
- **n:** n=2447 patient tumors (clinical-molecular layer); organoid cohort with WT, G12D, G12R lines (n=3-6 per genotype)

- **Effect size:** moderate — KrasG12R PDAC organoids showed decreased migration and improved survival in orthotopic models versus G12D, alongside a distinctive low-ERK / immune-enriched transcriptional signature. The G12R subset's lower baseline MAPK flux predicts a wider therapeutic window for active-state RAS inhibition than the high-flux G12D class.
- **Variance:** vs KRAS G12D and KRAS wild-type organoids; orthotopic injection comparators; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Tumor-microenvironment claim rests partly on bulk-RNA deconvolution, not single-cell validation across all samples. The lower-ERK phenotype is a relative not absolute observation.

Revolution Medicines/Revolution Medicines (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** zoldonrasib (RMC-7977) + ivonescimab; daraxonrasib + ivonescimab combinations
- **Indication / model:** Advanced / metastatic solid tumors with RAS mutation (NSCLC, CRC; PDAC eligibility per protocol) (2L+); RAS mutation; RAS(ON) inhibitor backbone with ivonescimab PD-1×VEGF bispecific
- **Design:** Open-label dose-exploration and expansion combining RAS(ON) inhibitors with ivonescimab
- **Effect size:** — — Safety, RP2D, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Astellas/Astellas (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ASP5834 pan-KRAS inhibitor (intravenous)
- **Indication / model:** Locally advanced / metastatic solid tumors with KRAS mutations or amplification — NSCLC, PDAC, CRC (2L+); KRAS G12V, G12D, G12C, G12R, G12A, G13D, or KRAS amplification (copy number ≥4)
- **Design:** Phase 1 IV dose-escalation + dose-expansion; combination with panitumumab in CRC
- **n:** 364
- **Effect size:** — — Safety, RP2D, ORR
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Verastem/Verastem (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** SBRT + avutometinib + defactinib
- **Indication / model:** Borderline resectable / locally advanced PDAC (neoadj); —
- **Design:** Randomized phase 2 (5:1) with SBRT plus FAK / RAF-MEK inhibition
- **n:** 36
- **Effect size:** — — Resectability, pathologic response, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

ADOPT investigators/ADOPT investigators (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Therapy from a menu including abemaciclib (CDK4/6), KRAS-directed agents, PARPi where indicated
- **Indication / model:** Advanced PDAC selected by patient-derived organoid (PDO) drug sensitivity (2L+); PDO sensitivity-based selection; genomic profiling overlay
- **Design:** Adaptive organoid-based precision-therapy platform
- **Effect size:** — — ORR, PFS, organoid-clinical concordance
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Incyte/Incyte (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** INCB161734 (KRAS G12D-selective) + chemotherapy
- **Indication / model:** Previously untreated KRAS G12D-mutated metastatic PDAC (DAWN-303) (1L); KRAS G12D
- **Design:** Randomized double-blind phase 3
- **Effect size:** — — OS, PFS
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Astellas/Astellas (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ASP3082 (setidegrasib, KRAS G12D-selective degrader) + mFOLFIRINOX or NALIRIFOX
- **Indication / model:** First-line KRAS G12D-mutated metastatic PDAC (1L); KRAS G12D
- **Design:** Phase 3 randomized
- **Effect size:** — — OS, PFS
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Joyo Pharmatech/Joyo Pharmatech (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** JYP0015 (ERAS-0015) oral pan-RAS molecular glue
- **Indication / model:** Advanced solid tumors with RAS mutation — PDAC, NSCLC, CRC cohorts (2L+); RAS mutation (pan-RAS molecular glue; KRAS G12X including G12R)
- **Design:** Open-label dose-exploration and expansion
- **Effect size:** — — Safety, RP2D, ORR, PFS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Sidney Kimmel Comprehensive Cancer Center/Sidney Kimmel Comprehensive Cancer Center (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** mutant-KRAS peptide vaccine (Poly-ICLC adjuvant) + balstilimab (anti-PD-1) + botensilimab (Fc-enhanced anti-CTLA-4)
- **Indication / model:** Stage IV MMR-proficient PDAC and colorectal cancer (2L+); Tumor-expressed KRAS mutation matching the vaccine panel; MMR-proficient
- **Design:** Phase 1 open-label, cohort-based
- **Effect size:** — — Safety, immunogenicity, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Pant/O'Reilly (2024) — Nature Medicine

Reference:PMID 38195752 · **Type:** clinical · **Inclusion:** included

- **Intervention:** ELI-002 2P (amphiphile mKRAS peptide vaccine)
- **Indication / model:** Resected PDAC and CRC with ctDNA / tumor-marker MRD positivity after locoregional therapy (AMPLIFY-201) (adj); MRD-positive after curative-intent surgery and standard adjuvant therapy; 25 patients (20 PDAC, 5 CRC); KRAS G12D or G12R required. The patient's G12R is in scope, but the patient is overt-recurrent rather than MRD-only.
- **Design:** Phase 1 open-label single-arm
- **n:** 25
- **Effect size:** 84 % with mKRAS-specific T-cell response — Safety, immunogenicity (T-cell response), tumor-biomarker response, relapse-free survival
- **Toxicities:** dose-limiting toxicity G>=3 (0/25, 0%); injection-site reaction (n=25)
- **Case fit:** partial
- **Notes:** G12R explicitly in the 2-peptide panel. Cohort is MRD-positive resected disease, not overt-recurrent — case fit is partial. Predecessor to ELI-002 7P (NCT05726864).

Krebs/Krebs (2024) — Journal of Clinical Oncology (ASCO 2024 abstract 4140)

Reference:doi:10.1200/JCO.2024.42.16_suppl.4140 · **Type:** clinical · **Inclusion:** included

- **Intervention:** avutometinib + defactinib + gemcitabine/nab-paclitaxel
- **Indication / model:** Treatment-naive metastatic PDAC, KRAS-mutant or wild-type (RAMP 205) (1L); Phase 1b/2; 41 patients dosed across 4 dose levels as of May 2024; KRAS activating mutation eligibility includes G12X / G13 / Q61. 1L untreated metastatic — patient is recurrent post-adjuvant FOLFIRI, so line fit depends on whether adjuvant FOLFIRI counts as prior systemic exposure in this protocol.
- **Design:** Phase 1b/2 open-label dose-finding plus expansion
- **n:** 41
- **Effect size:** 83 % ORR at dose level 1 (5/6) — Confirmed ORR (early signal at dose level 1)
- **Toxicities:** Manageable across four dose cohorts; details pending full publication. FDA orphan drug designation granted on the back of this signal.
- **Case fit:** partial
- **Notes:** ASCO 2024 abstract; full publication pending. 5-of-6 confirmed PRs at dose level 1 is striking but small.

RAMP 205 trial registry NCT05669482 corresponds to the trials.jsonl row; included here for the efficacy preview that would weight any avutometinib/defactinib-backbone discussion.

Ruijin Hospital / Abogen/— (2024)

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** ABO2102 (KRAS neoantigen mRNA vaccine) +/- toripalimab
- **Indication / model:** KRAS-mutated advanced solid tumors — PDAC and NSCLC cohorts (2L+); Tumor must carry one of five encoded KRAS-mutant neoantigens; whether G12R is among the five is not stated in the registry record. ECOG 0-2
- **Design:** Early phase 1 open-label, monotherapy and PD-1-combination arms (dose escalation + expansion)
- **Effect size:** — — safety / immunogenicity (ongoing)
- **Toxicities:** No clinical data released; preclinical only.
- **Case fit:** partial
- **Notes:** FDA IND May 2025; no clinical efficacy or safety data yet, so toxicities[] is empty (no source AE table exists). Encodes five KRAS-mutant antigens with dual CD8/CD4 activation; G12R inclusion in the panel is unconfirmed in the registry. MSS/low-TMB context makes a KRAS-directed vaccine plus PD-1 more rational than checkpoint monotherapy.

Wang/Shen (2024) — Cell Research

Reference:PMID 38914844 · **Type:** clinical · **Inclusion:** included

- **Intervention:** KRAS G12V mRNA vaccine + pembrolizumab (class precedent)
- **Indication / model:** Advanced KRAS G12V-mutant solid tumors (locally advanced PDAC; metastatic NSCLC) (2L+); HLA-A*11:01-restricted, KRAS G12V-mutant; 2-patient case series (one PDAC, one NSCLC)
- **Design:** Case series / correspondence
- **n:** 2
- **Effect size:** PR in 2/2 patients — objective response (RECIST 1.1)
- **Toxicities:** fever (2/2, 100%); injection-site pain (1/2, 50%)
- **Case fit:** cross_tumor_only
- **Notes:** Class precedent for the mRNA-KRAS-vaccine + PD-1 axis, grouped with ABO2102. Covers G12V only — not the patient's G12R allele — so it anchors mechanism, not allele-matched benefit; a 2-patient correspondence, not a trial.

Holderfield/Singh (2024) — Nature

Reference:PMID 38589574 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236)
- **Indication / model:** Pan-cancer cell-line panel (>700 lines) with KRAS / NRAS / HRAS variants including KRAS G12R; subcutaneous xenografts across G12D, G12V, G12R, G12C, Q61 backgrounds
- **Design:** Noncovalent tri-complex with cyclophilin A that binds active RAS-GTP and sterically blocks RAS-effector engagement across mutant and wild-type KRAS, NRAS, HRAS. Broad RAS(ON) blockade is the relevant mechanism for KRAS G12R, which is GTP-loaded and outside the reach of GDP-state covalent G12C drugs.
- **n:** in-vitro: hundreds of lines; in-vivo: standard xenograft cohorts (n not enumerated in abstract)

- **Effect size:** strong — RMC-6236 inhibited proliferation across G12X-mutant lines including G12R-bearing pancreatic and lung models, with on-target ERK suppression at sub-100 nM. Xenograft regression was achieved across multiple G12 alleles at tolerated oral exposures, and the molecule maintained activity in lines where adagrasib or sotorasib had no effect.
- **Variance:** vs vehicle and allele-selective comparators (e.g. sotorasib in G12C lines); translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** G12R-specific subgroup not separately quantified; activity is read from pooled G12X panels and a small number of G12R lines. Toxicology of pan-RAS blockade rests on a residual wild-type-RAS pharmacology window that is tighter in some tissues than others.

Wasko/Olive (2024) — Nature

Reference:PMID 38588697 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** zoldonrasib-class tool compound RMC-7977 (RAS(ON) multi-selective)
- **Indication / model:** PDAC translational suite: human and murine cell lines, patient-derived organoids, human PDAC explants, subcutaneous and orthotopic CDX / PDX, syngeneic allografts, and the autochthonous KPC (KrasLSL-G12D; Trp53LSL-R172H; Pdx1-Cre) GEMM
- **Design:** Active-state RAS(ON) tri-complex inhibitor selective for GTP-bound mutant and wild-type KRAS, NRAS, HRAS. Translates to PDAC because virtually all PDAC tumor cells depend on RAS-GTP signaling, and the differential tolerance reflects oncogene addiction rather than allele identity.
- **n:** Multiple model classes, n per arm varies (e.g. KPC mice n approx 10-20 per arm)
- **Effect size:** strong — Treated PDAC tumors showed waves of apoptosis with sustained proliferative arrest, while normal tissues had only transient and apoptosis-free decreases in proliferation. In autochthonous KPC mice RMC-7977 produced the longest survival extension recorded for that model and complete radiologic responses in a subset, with relapse linked to MYC copy-number gain.
- **Variance:** vs vehicle; chemotherapy comparators in some arms; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** KPC carries G12D rather than G12R, so the GEMM data are class-level rather than allele-specific. RMC-7977 is the tool compound; daraxonrasib (RMC-6236) is the clinical translate.

Jiang/Singh (2024) — Cancer Discovery

Reference:PMID 38593348 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236) translational evaluation
- **Indication / model:** Translational pan-RAS panel: cell lines, patient-derived organoids, and CDX / PDX models stratified by KRAS G12 allele (G12C, G12D, G12V, G12R) across PDAC, NSCLC, and CRC
- **Design:** RAS(ON) multi-selective inhibition with cyclophilin-A-dependent tri-complex. The translational companion to NCT05379985 characterizes activity stratified by G12 variant, including the rarer G12R allele relevant to this case.
- **n:** n not enumerated in abstract; multi-model translational series
- **Effect size:** strong — RMC-6236 produced dose-dependent ERK pathway suppression and tumor regression across G12C, G12D, G12V, and G12R-mutant PDAC models, with G12R-bearing lines responding within the same exposure window as other G12 alleles. Pharmacodynamic engagement (DUSP6, SPRY suppression) and tumor regression were concordant across alleles.

- **Variance:** vs vehicle; allele-selective KRAS inhibitors for cross-comparison; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** G12R cohort within the translational panel is smaller than G12D / G12V cohorts; allele-resolved durability data are read from a limited number of lines.

Principe/Grippo (2024) — JCI Insight

Reference:PMID 39288262 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** NIS793 / TGF-beta-1+2 antibody plus chemo + anti-PD-1 (preclinical rationale)
- **Indication / model:** Orthotopic and autochthonous murine PDAC models (KPC, KPP) with TGF-beta blockade by 1D11 (murine surrogate for NIS793)
- **Design:** TGF-beta drives carcinoma-associated fibroblast activation, ECM deposition, and T-regulatory cell recruitment that together build the immune-exclusion barrier in PDAC. Antibody blockade of TGF-beta-1 and TGF-beta-2 reverses the stromal program and re-permits drug penetration and T-cell entry.
- **n:** Multiple model groups; n=8-12 mice per arm
- **Effect size:** moderate — TGF-beta blockade reduced fibroblast activation and ECM deposition in metastatic PDAC models, and added incremental tumor control on top of chemotherapy and anti-PD-1. The mechanism anchored the daNIS-2 / daNIS-3 NIS793 phase 3 design.
- **Variance:** vs Vehicle; gem/nab-paclitaxel; anti-PD-1 alone; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Novartis discontinued NIS793 PDAC development in 2023 after the daNIS-2 phase 3 readout did not meet endpoints; the preclinical signal did not survive clinical translation. Included as a foundational mechanism reference for the CAF / stroma-targeting axis the board may revisit with next-generation TGF-beta-trap or LRRC15-targeting agents.

Revolution Medicines/Revolution Medicines (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236) monotherapy
- **Indication / model:** Previously treated metastatic pancreatic ductal adenocarcinoma (2L+); RAS mutation (KRAS / NRAS / HRAS codon 12/13/61); G12R within scope
- **Design:** Randomized, open-label, phase 3 versus investigator's choice SoC
- **Effect size:** — — OS, PFS, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Revolution Medicines/Revolution Medicines (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236) + chemo (mFOLFIRINOX or gemcitabine/nab-paclitaxel); RMC-9805 in G12D-specific subprotocols
- **Indication / model:** GI solid tumors (PDAC, CRC) with RAS mutation (1L and 2L+); RAS mutation; subprotocols for G12D / G12X / G12C variants

- **Design:** Open-label multi-arm platform of RAS(ON) inhibitors as monotherapy and with chemotherapy
- **Effect size:** — — Safety, RP2D, ORR, PFS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Pfizer/Pfizer (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** PF-07934040 pan-KRAS small-molecule inhibitor; combinations with gem/nab, FOLFOX, cetuximab, pembrolizumab
- **Indication / model:** Advanced solid tumors with KRAS mutations (G12C, G12D, G12V, G12R, G12S, G13D, Q61H) — NSCLC, CRC, PDAC (2L+ (mono); 1L (with gem/nab combo arm)); KRAS pan-G12 / G13D / Q61H, including G12R
- **Design:** Phase 1 open-label dose-escalation + dose-expansion monotherapy and combinations
- **n:** 330
- **Effect size:** — — Safety, RP2D, ORR, PFS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Pant/O'Reilly (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ELI-002 7P (lipid-conjugated amphiphile KRAS peptide vaccine + immune-stimulatory oligonucleotide)
- **Indication / model:** Resected PDAC and other KRAS / NRAS mutant solid tumors in the adjuvant minimal residual disease setting (adj); KRAS / NRAS G12D, G12R, G12V, G12A, G12C, G12S, G13D — 7-peptide panel
- **Design:** Phase 1/2 with adjuvant MRD-anchored design and randomized phase 2 expansion (2:1)
- **n:** 135
- **Effect size:** — — Recurrence-free survival, ctDNA reduction, immunogenicity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Academic consortium/Academic consortium (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MEK inhibitor + hydroxychloroquine (NTO-RAS)
- **Indication / model:** RAS-mutant solid tumors including PDAC (2L+); RAS mutation
- **Design:** Phase 2 multi-cohort
- **Effect size:** — — ORR, PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Amgen/Amgen (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** anvumetostat (AMG 193 / PRMT5 inhibitor) + gem/nab or mFOLFIRINOX or RMC-6236
- **Indication / model:** Advanced GI / biliary / pancreatic cancer with homozygous MTAP deletion (1L and 2L+); Homozygous MTAP deletion (commonly co-deleted with CDKN2A)
- **Design:** Phase 1b combination
- **Effect size:** — — Safety, RP2D, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Jacobio Pharmaceuticals/Jacobio Pharmaceuticals (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** JAB-23E73 oral pan-KRAS (ON/OFF) inhibitor
- **Indication / model:** Advanced solid tumors harboring KRAS gene alteration (2L+); KRAS alteration; pan-KRAS(ON/OFF) covering G12X (G12D/V/R/S/A), G13D, Q61H
- **Design:** Phase 1/2a open-label dose-escalation + expansion
- **Effect size:** — — Safety, RP2D, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Ruijin Hospital/Ruijin Hospital (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ABO2102 KRAS neoantigen mRNA vaccine ± toripalimab
- **Indication / model:** Advanced KRAS-mutated solid tumors — PDAC, NSCLC and others (2L+); KRAS mutation matching the vaccine's encoded neoantigen panel
- **Design:** Open-label monotherapy and PD-1-combination arms
- **Effect size:** — — Safety, immunogenicity, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Strickler/Hong (2023) — New England Journal of Medicine

Reference:PMID 36546651 · **Type:** clinical · **Inclusion:** included

- **Intervention:** sotorasib (AMG 510)
- **Indication / model:** Pretreated advanced KRAS p.G12C-mutated pancreatic ductal adenocarcinoma (CodeBreaK 100 PDAC cohort) (2L+); 38 patients with KRAS G12C-mutated PDAC; median 2 prior lines (range 1-8). G12C is a different allele from the patient's G12R — sotorasib does not bind G12R.
- **Design:** Phase 1/2 single-arm pooled (phase 1 n=12 + phase 2 n=26)
- **n:** 38
- **Effect size:** 21 % — ORR (centrally reviewed)
- **Variance:** 95% CI 10–37
- **Toxicities:** any treatment-related AE (16/38, 42%); any treatment-related AE G>=3 (6/38, 16%); diarrhea (n=38); fatigue (n=38)
- **Case fit:** cross_tumor_only
- **Notes:** Mechanism-class context only — sotorasib is G12C-selective and does not act on KRAS G12R. The

21% ORR establishes proof-of-concept that allele-selective covalent KRAS blockade works in PDAC, which is what anchors the pan-RAS / pan-G12X rationale for a G12R patient. Tag: case_match cross_tumor_only flags the mutation-class mismatch even though the indication matches.

Bekaii-Saab/Pant (2023) — Journal of Clinical Oncology

Reference:PMID 37099736 · **Type:** clinical · **Inclusion:** included

- **Intervention:** adagrasib (MRTX849)
- **Indication / model:** Advanced solid tumors with KRAS G12C mutation, excluding NSCLC and CRC (KRYSTAL-1 cohort) (2L+); 63 treated patients; pancreatic n=21, biliary tract n=12, other GI / gyne. G12C-only — not relevant to the patient's G12R at the molecular level.
- **Design:** Phase 1/2 single-arm cohort
- **n:** 63
- **Effect size:** 35.1 % — ORR per investigator
- **Toxicities:** any treatment-related AE (n=63, 96.8%); any treatment-related AE G>=3 (n=63, 27%); nausea (n=63, 49.2%); diarrhea (n=63, 47.6%); fatigue (n=63, 41.3%); vomiting (n=63, 39.7%)
- **Case fit:** cross_tumor_only
- **Notes:** PDAC subset ORR 33.3% (7/21) and biliary 41.7% (5/12) — consistent with sotorasib in the same indication. G12C-selective; off-axis for a G12R patient. Logged for the allele-selective-KRAS class precedent that the pan-RAS programs are extending to non-G12C alleles.

Wainberg/O'Reilly (2023) — Lancet

Reference:PMID 37708904 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 1L chemotherapy that does not target the patient's molecular features. Out-of-scope per Libby's feature-targeting rule.

- **Intervention:** NALIRIFOX (NAPOLI 3)
- **Indication / model:** Treatment-naive metastatic pancreatic adenocarcinoma, first-line
- **Design:** Randomized open-label phase 3 vs gem/nab
- **Case fit:** none
- **Notes:** Excluded: Standard-of-care 1L chemotherapy that does not target the patient's molecular features. Out-of-scope per Libby's feature-targeting rule. — OS 11.1 vs 9.2 mo (HR 0.83, p=0.036) — new 1L reference regimen for mPDAC. The patient has had FOLFIRI adjuvant (irinotecan-exposed) but is oxaliplatin-naive, which keeps this regimen on the SoC table at the treating team's discretion.

Smith/Christensen (2023) — Cancer Discovery

Reference:PMID 37552839 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** MRTX1719 (MTA-cooperative PRMT5 inhibitor)
- **Indication / model:** HCT116 isogenic MTAP-WT vs MTAP-deleted; broad cell-line panel of MTAP-deleted vs MTAP-WT tumors; xenograft models including MTAP-deleted PDAC, NSCLC, melanoma
- **Design:** Selective binding to the PRMT5-MTA complex, leveraging the elevated MTA pool in MTAP-deleted cells. The cooperative mechanism produces >70-fold selectivity for MTAP-deleted over MTAP-proficient cells, sparing normal MTAP-positive tissue and avoiding the dose-limiting hematologic toxicity of first-generation PRMT5 inhibitors.
- **n:** >300 cell lines in viability panel; in-vivo cohorts n=5-10 per arm across multiple xenografts

- **Effect size:** strong — MRTX1719 inhibited PRMT5-dependent SDMA methylation and cell viability in MTAP-deleted lines with >70-fold selectivity, and produced tumor regression in MTAP-deleted xenografts at well-tolerated doses. Hematologic and bone-marrow effects, the limiting feature of first-generation PRMT5 inhibitors, were not observed at efficacious doses.
- **Variance:** vs MTAP-WT matched lines; vehicle in xenografts; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** PDAC-specific xenografts represented a minority of the in-vivo panel; the bulk of in-vivo data is in NSCLC, mesothelioma, and melanoma models. Clinical translate for this case requires confirmed MTAP co-deletion.

Knudsen/Witkiewicz (2023) — Cancer Research

Reference:PMID 36346366 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CDK4/6 inhibition plus MEK / ERK inhibition (palbociclib + trametinib or ulixertinib)
- **Indication / model:** KRAS-mutant PDAC cell lines (MIA PaCa-2, Panc-1, AsPC-1, Capan-1) and matched patient-derived organoids; subcutaneous xenografts of KRAS-mutant PDAC
- **Design:** KRAS-driven ERK signaling and CDKN2A loss converge on RB inactivation. Concurrent CDK4/6 and ERK / MEK blockade cooperatively prevents the compensatory upregulation of cyclin D, MYC, and anti-apoptotic signaling that limits single-agent palbociclib in PDAC.
- **n:** Six PDO models; in-vivo cohorts n=8-12 per arm
- **Effect size:** strong — Combined CDK4/6 and ERK / MEK inhibition produced synergistic suppression of proliferation in all six PDAC organoid models tested and tumor regression in xenografts where either agent alone produced cytostasis. The combination anchored the perioperative palbociclib + binimetinib trial (NCT04870034) and the ulixertinib + palbociclib study (NCT03454035).
- **Variance:** vs vehicle; single-agent palbociclib; single-agent trametinib or ulixertinib; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Most lines are KRAS G12D / G12V; G12R-specific organoid data are absent. The synergy logic still applies because both axes converge on RB through the same effector chain, but G12R's lower baseline ERK flux is an open variable.

Verastem/Verastem (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** avutometinib (VS-6766, RAF/MEK clamp) + defactinib (FAKi) + gemcitabine + nab-paclitaxel
- **Indication / model:** Previously untreated metastatic PDAC with KRAS activating mutation (1L); KRAS activating mutation (any G12X including G12R; G13D; Q61)
- **Design:** Phase 1b/2a open-label
- **n:** 40
- **Effect size:** — — Safety, RP2D, ORR, PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Mirati/Mirati (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MRTX1133 (KRAS G12D-selective)
- **Indication / model:** Advanced solid tumors with KRAS G12D mutation (2L+); KRAS G12D
- **Design:** Phase 1/2 (phase 2 not initiated)
- **n:** 63
- **Effect size:** Terminated for formulation issues — Safety, RP2D, ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Renouf/O'Callaghan (2022) — Nature Communications

Reference:PMID 36028483 · **Type:** clinical · **Inclusion:** included

- **Intervention:** durvalumab + tremelimumab + gemcitabine/nab-paclitaxel
- **Indication / model:** Treatment-naive metastatic PDAC (CCTG PA.7) (1L); 180 randomized to chemo + dual ICI vs chemo alone; unselected for biomarker. The patient is recurrent rather than 1L untreated, but the negative readout is decision-relevant for any ICI-combination discussion.
- **Design:** Randomized open-label phase 2
- **n:** 180
- **Effect size:** — — Overall survival (primary)
- **Variance:** p=0.72
- **Toxicities:** lymphocyte elevation (n=180)
- **Case fit:** weak
- **Notes:** Negative RCT for dual ICI added to chemo in unselected metastatic PDAC. The subgroup signal in KRAS-wildtype patients is hypothesis-generating only. For an MSS / sub-10-TMB G12R patient, this is the load-bearing negative evidence for adding biomarker-agnostic ICI to chemo. — (primary endpoint not extracted)

Abrams/McCubrey (2022) — Cells

Reference:PMID 35269414 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** APR-246 / eprenetapopt (mutant TP53 reactivator)
- **Indication / model:** PDAC cell lines PANC-28 (TP53-null) and MIA PaCa-2 (TP53 R248W gain-of-function); isogenic wild-type TP53 reintroduction
- **Design:** APR-246 covalently modifies cysteine residues in mutant p53 to restore wild-type conformation and DNA-binding activity. The mechanism is selective for missense (especially conformationally unstable hotspot) mutants; pure loss-of-function or truncating TP53 variants lack a target to reactivate.
- **n:** n=3 biological replicates per condition
- **Effect size:** moderate — APR-246 sensitized MIA PaCa-2 (R248W gain-of-function) PDAC cells to combination cytotoxics and natural-compound derivatives, with effect size dependent on the presence of mutant or reintroduced wild-type TP53 protein. TP53-null PANC-28 cells were not sensitized, confirming target-class specificity.
- **Variance:** vs Vehicle; matched parental lines without WT-TP53 reintroduction; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** In-vitro only; no PDAC xenograft data. Activity requires a missense / conformationally unstable mutant TP53 protein. The case's TP53 variant class is annotated as inactivating without missense specification;

APR-246 is off-target if the variant is a truncation or splice-site allele.

Hallin/Christensen (2022) — Cancer Discovery

Reference:PMID 36216931 · **Type:** preclinical · **Inclusion:** excluded — G12D-selective by design; does not bind G12R. Reviewed because the preclinical PDAC efficacy is striking (deep regressions in implantable and autochthonous KPC) but the allele specificity means the mechanism is off-target for this patient. Logged here so the audit trail captures that the class precedent was examined and rejected on allele grounds; the pan-RAS RMC-6236 / RMC-7977 rows are the on-axis evidence for G12R.

- **Intervention:** MRTX1133 (KRAS G12D-selective small-molecule inhibitor)
- **Toxicities:** n/a (preclinical)
- **Case fit:** none
- **Notes:** Excluded: G12D-selective by design; does not bind G12R. Reviewed because the preclinical PDAC efficacy is striking (deep regressions in implantable and autochthonous KPC) but the allele specificity means the mechanism is off-target for this patient. Logged here so the audit trail captures that the class precedent was examined and rejected on allele grounds; the pan-RAS RMC-6236 / RMC-7977 rows are the on-axis evidence for G12R.

Revolution Medicines/Revolution Medicines (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236) oral monotherapy
- **Indication / model:** Advanced solid tumors with KRAS/NRAS/HRAS mutations at codons 12, 13, or 61 — includes PDAC, NSCLC, CRC cohorts (2L+); KRAS G12X (including G12R), NRAS, HRAS at codons 12/13/61; PDAC RAS-WT also eligible
- **Design:** Multicenter open-label dose-escalation + dose-expansion (FIH)
- **Effect size:** — — Safety, RP2D, ORR (RECIST 1.1), PFS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Bekaii-Saab/Bekaii-Saab (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** trametinib 2 mg/day + hydroxychloroquine 1200 mg/day (PaTcH)
- **Indication / model:** Metastatic refractory pancreatic cancer (2L+); —
- **Design:** Single-arm phase 2
- **n:** 20
- **Effect size:** Terminated for futility — ORR, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Maitra/Maitra (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** MEK inhibitor-based combinations (MEKi + HCQ, MEKi + EGFRi, etc.)
- **Indication / model:** Advanced KRAS G12R altered PDAC (2L+); KRAS G12R
- **Design:** Observational registry of MEK-based combination therapy
- **Effect size:** — — ORR, PFS, OS (observational)
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Reiss/Domchek (2021) — Journal of Clinical Oncology

Reference:PMID 33970687 · **Type:** clinical · **Inclusion:** included

- **Intervention:** rucaparib (Rubraca)
- **Indication / model:** Platinum-sensitive advanced PDAC with germline or somatic BRCA1/2 or PALB2 pathogenic variant (maintenance); 46 enrolled, 42 evaluable; germline BRCA1 n=7, BRCA2 n=27, PALB2 n=6; somatic BRCA2 n=2. Broader than POLO by including PALB2 and somatic variants.
- **Design:** Phase 2 single-arm
- **n:** 46
- **Effect size:** 59.5 % PFS rate at 6 months — PFS at 6 months (primary), ORR, OS
- **Toxicities:** anemia (n=46, 74%); anemia G>=3 (n=46, 22%); nausea (n=46, 48%); nausea G>=3 (n=46, 2%); increased ALT (n=46, 47%); increased ALT G>=3 (n=46, 4%); fatigue (n=46, 45%); fatigue G>=3 (n=46, 4%); thrombocytopenia (n=46, 39%); thrombocytopenia G>=3 (n=46, 4%); dysgeusia (n=46, 37%)
- **Case fit:** partial
- **Notes:** Extends the PARP signal beyond gBRCA to include sBRCA and PALB2; relevant for the case only if germline / somatic HRD testing returns positive. Currently not on file; gated by the germline panel in target_validation.jsonl. Safety data extracted from in-text Safety paragraph and Appendix Table A4 of JCO 2021;39:2497-2505.

Raybould/McRee (2021) — Journal of Clinical Oncology (ASCO abstract 3103)

Reference:doi:10.1200/JCO.2021.39.15_suppl.3103 · **Type:** clinical · **Inclusion:** included

- **Intervention:** ulixertinib (ERK1/2i) + palbociclib (CDK4/6i)
- **Indication / model:** Advanced solid tumors (13 CRC, 9 PDAC, 2 melanoma, 1 cholangiocarcinoma, 1 GIST) (2L+); MAPK-pathway-altered advanced solid tumors; nine pancreatic patients in the escalation. 16 of 26 evaluable for response
- **Design:** Phase 1b dose-escalation, open-label single-arm
- **n:** 26
- **Effect size:** 2 SD (PDAC, cholangiocarcinoma) — disease control (stable disease)
- **Toxicities:** fatigue (n=26, 70%); rash (n=26, 62%); nausea (n=26, 54%); lymphocyte count decreased (n=26, 77%); white cell count decreased (n=26, 73%); anemia (n=26, 65%); white cell count decreased G3 (n=26); lymphocyte count decreased G3 (n=26); anemia G3 (n=26); fatigue G3 (n=26)
- **Case fit:** partial
- **Notes:** ASCO 2021 abstract-only (no full paper); per-term grade-3 rates not tabulated in the abstract, so those toxicity rows carry the term without a rate. Hits two patient features at once — ERK blockade downstream of KRAS G12R and CDK4/6 inhibition for CDKN2A loss / CCND3 — but the abstract enrolled by MAPK alteration and histology, not by G12R, and the pancreatic signal was stable disease, not response.

Rakhra/Irvine (2021) — Journal for ImmunoTherapy of Cancer

Reference:PMID 34376552 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Amphiphile mKRAS peptide vaccine (ELI-002 preclinical lead)
- **Indication / model:** C57BL/6 mice bearing syngeneic CT26 and MC38 colorectal allografts (KrasG12D) and Panc02-derived models; lymph-node-targeting amphiphile peptide constructs with CpG-7909 adjuvant
- **Design:** Albumin-binding lipid conjugation drives peptide trafficking to draining lymph nodes, where high-density antigen presentation and lymph-node-resident CpG signaling expand mKRAS-specific CD8 and CD4 T cells. The strategy targets the KRAS peptide neoantigens (G12D and G12R variants) shared across PDAC and CRC.
- **n:** n approx 8-10 mice per arm across efficacy experiments
- **Effect size:** strong — Amphiphile delivery generated tenfold-greater mKRAS-specific T-cell responses than soluble peptide and produced durable tumor control with cures in a fraction of KRAS-mutant syngeneic models. Memory T-cell expansion persisted for >100 days and conferred resistance to tumor rechallenge.
- **Variance:** vs Soluble (non-amphiphile) peptide vaccine; CpG alone; vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Murine syngeneic models are MSI-high relative to human PDAC and likely overstate immunogenic vaccine windows. The clinical ELI-002 AMPLIFY-201 readout in MRD-positive PDAC is the more conservative translatability anchor.

NCI/NCI (2021)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** olaparib maintenance
- **Indication / model:** Resected PDAC with germline or somatic BRCA1/2 or PALB2 mutation (adj); Germline / somatic BRCA1, BRCA2, or PALB2 pathogenic variant
- **Design:** Randomized double-blind phase 2 (APOLLO)
- **n:** 152
- **Effect size:** — — Relapse-free survival (primary)
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Marabelle/Diaz (2020) — Journal of Clinical Oncology

Reference:PMID 31682550 · **Type:** clinical · **Inclusion:** included

- **Intervention:** pembrolizumab (Keytruda)
- **Indication / model:** MSI-H / dMMR previously treated noncolorectal advanced cancer (KEYNOTE-158) (2L+); 233 enrolled across 27 tumor types; pancreatic n=22; MSI-H / dMMR required. The patient is MSS — biomarker-excluded from this indication. Logged to harden the MSS-PDAC ICI foreclosure call.
- **Design:** Multi-cohort open-label phase 2
- **n:** 233
- **Effect size:** 18.2 % ORR in MSI-H pancreatic subgroup (4/22) — ORR overall and per-tumor (pancreatic subset)
- **Toxicities:** any treatment-related AE G>=3 (34/233, 14.6%); discontinuation for immune-mediated AE (n=233, 5.2%)

- **Case fit:** none
- **Notes:** The PDAC ORR (18%) is real only for MSI-H disease, which the patient does not have. PDAC subgroup mPFS 2.0 / mOS 4.0 in the responder-enriched MSI-H cohort gives a ceiling for what monotherapy ICI delivers even when biomarker-positive. The patient is MSS / TMB 4.1 — the door is closed.

Hobbs/Der (2020) — Cancer Discovery

Reference:PMID 31649109 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** KRAS G12R signaling-bias characterization (informs pan-RAS targeting rationale)
- **Indication / model:** Isogenic RIE-1 rat intestinal epithelial cells and PDAC lines (HPAC, Capan-1) with knock-in of KRAS G12D vs G12V vs G12R; biochemical and structural characterization
- **Design:** Structural analysis showing the G12R arginine substitution partially unfolds helix alpha-2 of switch II, displacing Q61 and forming a direct interaction with E62 / T35. The resulting altered switch-II surface impairs PI3K p110-alpha binding and abolishes macropinocytosis induction relative to G12D / G12V, while preserving Raf engagement.
- **n:** Biochemical assays (n=3 biological replicates typical); structural data from x-ray crystallography
- **Effect size:** moderate — KRAS G12R is functionally distinct from other G12 alleles: stable expression of G12D or G12V increased macropinocytosis whereas G12R did not, and G12R-PDAC lines showed selective dependence on alternative scavenging pathways. The work anchors why allele-class context matters and why G12R sits in the RAS(ON) tri-complex window rather than the GDP-state covalent G12C window.
- **Variance:** vs KRAS wild-type and other G12 variants (G12D, G12V) as comparators; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Biology paper, not a therapeutic efficacy paper. Provides the mechanistic basis for selecting pan-RAS over covalent G12C-class drugs in this case; does not by itself predict on-treatment response magnitude.

Roberti/Pellegrini (2020) — Journal of Medicinal Chemistry

Reference:PMID 31790586 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** olaparib plus BRCA2-RAD51 disruptor (preclinical synthetic-lethal extension)
- **Indication / model:** Capan-1 (BRCA2-deficient) and BxPC-3 (BRCA2-proficient) PDAC cell lines; isogenic comparisons
- **Design:** PARP inhibition collapses single-strand-break repair, generating replication-fork lesions that require BRCA-dependent homologous recombination for resolution. BRCA1/2- or PALB2-deficient PDAC cells lack this repair channel, producing the synthetic-lethal vulnerability that drove the olaparib FDA approval in germline-BRCA-mutated PDAC.
- **n:** Standard biological triplicates
- **Effect size:** strong — Capan-1 (BRCA2-null) PDAC cells were >100-fold more sensitive to olaparib than BxPC-3 (BRCA2-wild-type) cells, recapitulating the canonical synthetic-lethal phenotype. Combining olaparib with a small-molecule BRCA2-RAD51 disruptor extended the sensitivity into otherwise-resistant lines.
- **Variance:** vs Vehicle; single-agent olaparib; single-agent BRCA2-RAD51 disruptor; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Mechanism is irrelevant to this case unless the germline panel returns a BRCA1, BRCA2, or PALB2 hit. Included as the mechanism-of-action reference behind the POLO and APOLLO trial design, which the board may pivot to if germline testing turns positive.

NCI/NCI (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** palbociclib
- **Indication / model:** Rare solid tumors with actionable genomic alteration — platform (2L+); CDKN2A mutation (palbociclib arm)
- **Design:** Genotyping-guided precision-medicine platform for rare tumors
- **Effect size:** — — ORR, clinical benefit
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Golan/Kindler (2019) — New England Journal of Medicine

Reference:PMID 31157963 · **Type:** clinical · **Inclusion:** included

- **Intervention:** olaparib (Lynparza)
- **Indication / model:** Germline BRCA1/2-mutated metastatic PDAC, maintenance after first-line platinum-based chemotherapy (POLO) (maintenance); 154 randomized; gBRCA1/2 pathogenic variant; no progression after ≥16 weeks of first-line platinum. The patient has no germline data yet and was treated with adjuvant FOLFIRI (no platinum exposure), so both biomarker and platinum-exposure gates are open.
- **Design:** Randomized double-blind placebo-controlled phase 3
- **n:** 154
- **Effect size:** 0.53 HR — Investigator-assessed PFS (primary)
- **Variance:** 95% CI 0.35–0.82; p=0.004
- **Toxicities:** any treatment-related AE G>=3 (n=92, 40%); any treatment-related AE G>=3 (n=62, 23%); discontinuation for AE (n=92, 5%)
- **Case fit:** partial
- **Notes:** Pivotal RCT driving FDA approval of olaparib maintenance in gBRCA-mutant metastatic PDAC. The case requires both germline testing (currently outstanding) and platinum induction (currently not on file) to unlock this pathway.

Bryant/Der (2019) — Nature Medicine

Reference:PMID 30833752 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ERK / MEK inhibition plus autophagy blockade (trametinib + chloroquine / hydroxychloroquine)
- **Indication / model:** Panel of KRAS-mutant PDAC cell lines (MIA PaCa-2, Pa01C, Pa02C, Pa14C, Pa16C); patient-derived xenografts; KPC GEMM-derived organoids and allografts
- **Design:** KRAS or ERK suppression drives compensatory autophagy that protects PDAC cells from metabolic stress. Combined ERK / MEK and autophagy blockade collapses both cytosolic and mitochondrial bioenergetic adaptation, producing synthetic lethal vulnerability that single-agent MEK or autophagy targeting does not deliver.
- **n:** in-vitro 3-6 biological replicates; in-vivo n approx 6-10 mice per arm
- **Effect size:** strong — Combined ERKi or trametinib with chloroquine produced near-complete tumor regression across PDAC cell line xenografts and patient-derived models, where single agents at best slowed growth. The combination also extended survival in autochthonous KPC mice and synergized in syngeneic CT26 KRAS-mutant CRC allografts.

- **Variance:** vs vehicle; single-agent SCH772984 (ERKi) or chloroquine alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Clinical translation has been mixed: PaTcH (trametinib + HCQ) terminated for futility despite the preclinical signal. The dose-intensity gap between mouse HCQ exposure and tolerable human dosing is the working hypothesis for the disconnect.

Kinsey/McMahon (2019) — Nature Medicine

Reference:PMID 30833748 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** MEK inhibition plus autophagy blockade (trametinib + HCQ; companion to Bryant 2019)
- **Indication / model:** KRAS-mutant PDAC cell lines; patient-derived xenografts; KPC GEMM; melanoma and CRC RAS / RAF-mutant lines for cross-tumor validation; one human case report of a metastatic PDAC patient treated with trametinib + HCQ
- **Design:** RAF-MEK-ERK inhibition induces a LKB1-AMPK-ULK1-dependent protective autophagy program that buffers metabolic stress in RAS-driven cancers. Simultaneous HCQ-mediated lysosomal disruption blocks this adaptive response and converts cytostasis into cytotoxicity.
- **n:** Multiple in-vitro lines; in-vivo cohorts of approx 5-10 mice per arm; human n=1 anecdotal case
- **Effect size:** strong — Combined MEK inhibition and autophagy blockade caused tumor regression in PDAC PDX and KPC models where either monotherapy delivered only growth slowing. A companion human case showed a metabolic response on FDG-PET and clinical benefit in a chemorefractory KRAS-mutant PDAC patient on trametinib plus HCQ.
- **Variance:** vs vehicle; single-agent trametinib; single-agent HCQ; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Companion paper to Bryant 2019 (back-to-back NEJM-style co-publication). Same caveat applies: PaTcH futility tempers the clinical signal; the autophagy axis still drives the NTO-RAS basket and several next-generation ULK1 / VPS34 trials.

Canon/Lipford (2019) — Nature

Reference:PMID 31666701 · **Type:** preclinical · **Inclusion:** excluded — G12C-selective covalent inhibitor; binds the cryptic GDP-state switch-II pocket via a reactive cysteine that G12R does not present. Off-target for this case. Reviewed because the preclinical foundation (anti-tumor immunity, immune-checkpoint synergy) is the class precedent the pan-RAS programs build on; the published efficacy is in NSCLC and CRC G12C models, not G12R PDAC. Logged for audit trail.

- **Intervention:** AMG 510 / sotorasib (KRAS G12C-selective covalent inhibitor)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: G12C-selective covalent inhibitor; binds the cryptic GDP-state switch-II pocket via a reactive cysteine that G12R does not present. Off-target for this case. Reviewed because the preclinical foundation (anti-tumor immunity, immune-checkpoint synergy) is the class precedent the pan-RAS programs build on; the published efficacy is in NSCLC and CRC G12C models, not G12R PDAC. Logged for audit trail.

CCTG/CCTG (2018)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** palbociclib (CDK4/6 inhibitor)
- **Indication / model:** Advanced solid tumors with actionable genomic alteration — basket (2L+); CDKN2A loss, CDK4 / CCND1 / SMARCA4 alteration (palbociclib arm — now closed)
- **Design:** Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR) — phase 2 basket
- **Effect size:** — — ORR, clinical benefit
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

UNC Lineberger/UNC Lineberger (2018)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ulixertinib (ERK1/2 inhibitor) + palbociclib (CDK4/6 inhibitor)
- **Indication / model:** Advanced pancreatic cancer and other solid tumors (2L+); KRAS/NRAS/HRAS-mutant MAPK activation (melanoma arm genotype-gated); PDAC expansion by histology
- **Design:** Phase 1 dose-escalation + PDAC/melanoma expansion cohorts
- **Effect size:** — — Safety, RP2D, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Aprea/Aprea (2017)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** eprenetapopt (APR-246) ± chemotherapy
- **Indication / model:** Platinum-resistant esophageal / gastroesophageal junction cancer with TP53 mutation (2L+); TP53 missense mutation (dominant-negative / GOF reactivation target)
- **Design:** Phase 1b/2 single-arm (APROC)
- **n:** 5
- **Effect size:** — — Safety, ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Wang-Gillam/Chen (2016) — Lancet

Reference:PMID 26615328 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 2L chemotherapy that does not target the patient's molecular features. Out-of-scope per Libby's feature-targeting rule. The patient already received FOLFIRI adjuvant, so prior irinotecan further limits the second-line nal-IRI option.

- **Intervention:** nanoliposomal irinotecan + 5-FU/LV (NAPOLI-1)
- **Indication / model:** Metastatic pancreatic adenocarcinoma, second-line after gemcitabine-based therapy
- **Design:** Randomized phase 3 three-arm
- **Case fit:** none
- **Notes:** Excluded: Standard-of-care 2L chemotherapy that does not target the patient's molecular features. Out-of-scope per Libby's feature-targeting rule. The patient already received FOLFIRI adjuvant, so prior irinotecan further limits the second-line nal-IRI option. — OS 6.1 vs 4.2 mo with nal-IRI+5FU vs 5FU, HR 0.67 — the comparator that the recurrent-PDAC SoC discussion would otherwise default to. Same scoping rationale as the other SoC chemo rows.

Jiang/DeNardo (2016) — Nature Medicine

Reference:PMID 27376576 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** FAK inhibition (VS-4718, defactinib precursor) plus checkpoint immunotherapy
- **Indication / model:** Autochthonous KPC mouse model of PDAC (Kras-LSL-G12D; Trp53-LSL-R172H; Pdx1-Cre); human PDAC tissue correlative analysis
- **Design:** Hyperactivated FAK in PDAC tumor cells drives a fibrotic, immunosuppressive stromal program that excludes T cells. Pharmacologic FAK inhibition collapses the stromal barrier, reduces MDSC and tumor-associated macrophage infiltration, and primes the tumor for checkpoint-blockade response.
- **n:** KPC cohorts n=8-15 per arm across multiple efficacy experiments; human tissue n>50 in correlative arm
- **Effect size:** strong — VS-4718 monotherapy doubled survival in KPC mice and reduced tumor fibrosis and MDSC infiltration; adding anti-PD-1 or anti-CTLA-4 produced sustained tumor regression in a model that is otherwise refractory to checkpoint blockade. The combination established the FAK + ICI rationale that drove the defactinib + pembrolizumab and chemo trials in PDAC.
- **Variance:** vs vehicle; anti-PD-1 or anti-CTLA-4 monotherapy; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** KPC is G12D, not G12R; immune-microenvironment biology may differ in the lower-ERK G12R context (Singhi 2025). The downstream clinical FAK + ICI combinations have shown modest signals at best in unselected PDAC.

Mavrakis/Sellers (2016) — Science

Reference:PMID 26912361 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRMT5 dependence in MTAP / CDKN2A co-deleted cancers
- **Indication / model:** Pan-cancer cell-line panel (>390 lines) including MTAP-deleted PDAC, NSCLC, and glioma lines; isogenic MTAP knockout and reconstitution clones
- **Design:** MTAP loss (co-deleted with CDKN2A on 9p21 in ~80-90% of CDKN2A homozygous deletions) drives intracellular accumulation of methylthioadenosine (MTA), which partially inhibits PRMT5. MTAP-deleted cells thus operate near a PRMT5 functional threshold, creating a synthetic-lethal vulnerability to further PRMT5 pharmacologic blockade.
- **n:** n=390+ lines; multiple isogenic clone replicates
- **Effect size:** strong — MTAP deletion sensitized cell lines to PRMT5 knockdown across tumor histologies, and reconstitution of MTAP rescued the dependence. The work established the molecular basis for the MTA-cooperative PRMT5 inhibitor programs (MRTX1719, AMG 193, BMS-986504) now in clinical development for MTAP-deleted PDAC and other indications.
- **Variance:** vs MTAP wild-type matched lines; MTAP-reconstituted MTAP-deleted lines; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Translates to this case only if MTAP is co-deleted with CDKN2A on the same allele; an MTAP IHC or NGS copy-number call is the gating measurement. First-generation PRMT5 inhibitors lacked selectivity; the clinical translate is the MTA-cooperative class (MRTX1719 etc.).

Kryukov/Stegmeier (2016) — Science

Reference:PMID 26912360 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRMT5 dependence in MTAP-deleted cancers (Kryukov companion paper)
- **Indication / model:** Project DRIVE / Achilles RNAi screens across hundreds of cancer cell lines; biochemical PRMT5 / MTA assays; MTAP-isogenic systems
- **Design:** Independently demonstrates that MTAP-deleted cancers accumulate MTA, which competes at the PRMT5 active site and reduces baseline PRMT5 methylation activity. The hypomorphic PRMT5 state is selectively sensitive to additional PRMT5 inhibition, the synthetic-lethal vulnerability that defines the MTA-cooperative inhibitor class.
- **n:** n>390 cell lines in genome-wide RNAi
- **Effect size:** strong — Genome-wide shRNA screening identified PRMT5 as the top differential dependency in MTAP-deleted lines across multiple tumor histologies. The discovery, published back-to-back with Mavrakis 2016, anchors the bilateral preclinical foundation for the MTAP / PRMT5 synthetic-lethal class.
- **Variance:** vs MTAP-proficient matched lines; in-vitro PRMT5 enzymatic assays with and without MTA; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Genetic dependence does not by itself indicate a tractable therapeutic window; first-generation PRMT5 inhibitors were too toxic, and the clinical translate required the MTA-cooperative chemical class to spare normal MTAP-proficient tissue.

Franco/Knudsen (2014) — Oncotarget

Reference:PMID 25156567 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** palbociclib (PD-0332991) plus pathway-selective combinations
- **Indication / model:** Panel of 15 patient-derived PDAC primary explants; PDX-derived organoid models; established cell lines with CDKN2A loss
- **Design:** CDKN2A loss derepresses CDK4/6, which would predict palbociclib sensitivity, but pancreatic biology drives parallel compensatory pathways (PI3K-mTOR, ERK feedback, IGF1R) that limit single-agent activity. Combination with pathway-selective agents collapses the bypass routes.
- **n:** n=15 PDAC explants; multiple cell-line replicates
- **Effect size:** moderate — Palbociclib suppressed proliferation in 14 of 15 primary PDAC explants in vitro, with the single resistant explant explained by RB1 loss. Single-agent durability in vivo was limited; combinations with PI3K, MEK, or IGF1R inhibitors converted cytostasis into sustained tumor control in PDX models.
- **Variance:** vs vehicle; single-agent PD-0332991; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Single-agent CDK4/6 inhibition has subsequently underperformed in CDKN2A-altered solid-tumor baskets (NCI-MATCH Z1C palbociclib ORR 4%; CDK4/6-amplified TAPUR PDAC arm). Translates as combination-strategy rationale rather than monotherapy support.

Heilmann/Knudsen (2014) — Cancer Research

Reference:PMID 24986516 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** palbociclib (PD-0332991) plus IGF1R inhibition (BMS-754807)
- **Indication / model:** p16INK4A (CDKN2A)-deficient PDAC cell lines (Panc1, MIA PaCa-2 derivatives, BxPC-3); subcutaneous xenografts
- **Design:** CDKN2A-deficient PDAC retains active IGF1R / mTORC1 signaling that drives

cyclin-D-CDK4/6-independent proliferation. IGF1R / IR co-inhibition collapses the mTORC1 escape route; substitution of an mTOR inhibitor (temsirolimus) broadens the sensitivity envelope.
- **n:** Multiple cell-line replicates; in-vivo cohorts n=8-10 per arm
- **Effect size:** strong — BMS-754807 plus PD-0332991 strongly suppressed proliferation in p16-deficient PDAC lines in-vitro and produced tumor regression in xenografts where either agent alone produced only cytostasis. Sensitivity correlated with mTORC1 suppression, and rapalog substitution recapitulated the synergy.
- **Variance:** vs vehicle; single-agent PD-0332991; single-agent BMS-754807; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Cell-line dominated panel without GEMM validation; IGF1R-class drugs have not delivered in clinical PDAC trials. The mechanism remains the cleanest rationale for CDK4/6 + mTOR / PI3K combinations in CDKN2A-loss PDAC.

Von Hoff/Renschler (2013) — New England Journal of Medicine

Reference:PMID 24131140 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care chemotherapy that does not target the patient's molecular features. Out-of-scope per Libby's feature-targeting rule. Logged as the comparator context for the gem/nab combination arms in RAMP-205 and RMC-GI-102.

- **Intervention:** nab-paclitaxel + gemcitabine (MPACT)
- **Indication / model:** Metastatic pancreatic adenocarcinoma, first-line
- **Design:** Randomized phase 3 vs gemcitabine
- **Case fit:** none
- **Notes:** Excluded: Standard-of-care chemotherapy that does not target the patient's molecular features. Out-of-scope per Libby's feature-targeting rule. Logged as the comparator context for the gem/nab combination arms in RAMP-205 and RMC-GI-102. — OS 8.5 vs 6.7 mo, HR 0.72 — the second 1L PDAC standard. Same scoping rationale as the FOLFIRINOX row.

Feig/Fearon (2013) — PNAS

Reference:PMID 24277834 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CXCL12 / CXCR4 blockade (AMD3100) plus anti-PD-L1 in PDAC
- **Indication / model:** Autochthonous KPC (Kras-LSL-G12D; Trp53-LSL-R172H; Pdx1-Cre) PDAC GEMM; FAP-DTR transgenic depletion of FAP-positive carcinoma-associated fibroblasts
- **Design:** FAP-positive carcinoma-associated fibroblasts produce CXCL12 that coats PDAC tumor cells and excludes CXCR4-expressing CD8 T cells, accounting for the canonical immune-cold phenotype. CXCR4 antagonism rapidly admits T cells to the tumor bed, where anti-PD-L1 then drives tumor regression.
- **n:** n approx 5-10 mice per arm across multiple experiments
- **Effect size:** strong — AMD3100 produced rapid T-cell accumulation in PDAC tumors and synergized with anti-PD-L1 to substantially reduce tumor burden, with effects lost when FAP-positive fibroblasts were depleted. The work established the CXCL12-CXCR4-FAP-CAF axis as the dominant immune-exclusion mechanism in KPC PDAC.
- **Variance:** vs Vehicle; anti-CTLA-4 monotherapy; anti-PD-L1 monotherapy; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Clinical translation has been disappointing: motixafortide + cemiplimab and other CXCR4-axis combinations have produced modest signals in MSS PDAC. KPC is G12D, not G12R; the G12R

immune-microenvironment phenotype may differ (Singhi 2025 suggests a baseline immune-enriched compartment).

Conroy/Ducreux (2011) — New England Journal of Medicine

Reference:PMID 21561347 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care chemotherapy for the indication whose mechanism does not target the patient's targetable features (KRAS G12R, CDKN2A loss, CCND3, MSS, TMB 4.1). Out-of-scope per Libby's feature-targeting rule. Included as a one-line audit-trail entry so the reviewer can see it was reviewed and decided not to compile.

- **Intervention:** FOLFIRINOX
- **Indication / model:** Metastatic pancreatic adenocarcinoma, first-line (PRODIGE 4/ACCORD 11)
- **Design:** Randomized phase 3 vs gemcitabine
- **Case fit:** none
- **Notes:** Excluded: Standard-of-care chemotherapy for the indication whose mechanism does not target the patient's targetable features (KRAS G12R, CDKN2A loss, CCND3, MSS, TMB 4.1). Out-of-scope per Libby's feature-targeting rule. Included as a one-line audit-trail entry so the reviewer can see it was reviewed and decided not to compile. — OS 11.1 vs 6.8 mo, HR 0.57 for FOLFIRINOX vs gemcitabine — landmark 1L PDAC benchmark. The treating team handles SoC chemo independent of Libby; logging only as a comparator-context anchor for the board's framing of what 'standard alternatives deliver'.

Olive/Tuveson (2009) — Science

Reference:PMID 19460966 · **Type:** preclinical · **Inclusion:** excluded — Foundational stroma-as-drug-delivery-barrier mechanism paper that motivated the entire FAP / CAF / stromal-targeting field. Reviewed but not retained as a primary entry because the hedgehog-pathway angle has not produced clinical translation (IPI-926 and vismodegib trials failed) and the more directly relevant preclinical anchors for this case are Jiang 2016 (FAK) and Feig 2013 (CXCL12). Logged to show the lineage was considered.

- **Intervention:** Olive 2009 Science / hedgehog-pathway stromal-vehicle PDAC paper
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Foundational stroma-as-drug-delivery-barrier mechanism paper that motivated the entire FAP / CAF / stromal-targeting field. Reviewed but not retained as a primary entry because the hedgehog-pathway angle has not produced clinical translation (IPI-926 and vismodegib trials failed) and the more directly relevant preclinical anchors for this case are Jiang 2016 (FAK) and Feig 2013 (CXCL12). Logged to show the lineage was considered.